

Differentials

If $z = f(x, y)$ then dx, dy and dz are called differentials in x, y and z respectively

Differential in z is given by

$$dz = \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy = \frac{\partial z}{\partial x} dx + \frac{\partial z}{\partial y} dy$$

Example

$$z = x^2 y$$

$$\begin{aligned} dz &= \frac{\partial z}{\partial x} dx + \frac{\partial z}{\partial y} dy \\ &= 2xy dx + x^2 dy \end{aligned}$$

1) Find differential in z if $f(x, y) = \frac{x^3}{3} - xy^2$

$$\frac{\partial f}{\partial x} = x^2 - y^2, \quad \frac{\partial f}{\partial y} = 0 - 2xy = -2xy$$

$$dz = \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy = (x^2 - y^2) dx - 2xy dy$$

Exact differential equation

A diff. eq of the form $M(x, y) dx + N(x, y) dy = 0$ is said to be an exact diff. eq if there exists a function $u(x, y)$ such that $\frac{\partial u}{\partial x} = M(x, y)$ and $\frac{\partial u}{\partial y} = N(x, y)$

$$\text{In this case } M dx + N dy = \frac{\partial u}{\partial x} dx + \frac{\partial u}{\partial y} dy$$

$$\text{or } M dx + N dy = du$$

This is called an exact differential

Its solution is given by $u(x, y) = c$

Example consider the diff. eq

$$(1-x) dx - (1+y) dy = 0, \quad M = 1-x, \quad N = -1-y$$

$$\text{Let } u(x, y) = \left(x - \frac{x^2}{2}\right) - \left(y + \frac{y^2}{2}\right)$$

then $\frac{\partial u}{\partial x} = 1-x = M$ $\frac{\partial u}{\partial y} = -(1+y) = N$

Given diff. eq is exact.

Solution is given by

$$u(x, y) = C \Rightarrow x - \frac{x^2}{2} - (y + \frac{y^2}{2}) = C$$

Condition for exactness

consider the diff. eq

$$M(x, y) dx + N(x, y) dy = 0 \text{ which is exact}$$

$\therefore$ There is a function $u(x, y)$ such that

$$\frac{\partial u}{\partial x} = M \quad \text{and} \quad \frac{\partial u}{\partial y} = N$$

$$\frac{\partial^2 u}{\partial y \partial x} = \frac{\partial M}{\partial y} \quad \text{and} \quad \frac{\partial^2 u}{\partial x \partial y} = \frac{\partial N}{\partial x}$$

$$\text{But } \frac{\partial^2 u}{\partial y \partial x} = \frac{\partial^2 u}{\partial x \partial y} \Rightarrow \frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

$\therefore$ A diff. eq of the form $M(x, y) dx + N(x, y) dy = 0$ is exact if $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$

Solution of an exact diff. eq.

$$\text{Solve } (x^2 - y^2) dx - 2xy dy = 0$$

$$M = x^2 - y^2, \quad N = -2xy$$

$$\frac{\partial M}{\partial y} = -2y \quad \frac{\partial N}{\partial x} = -2y$$

$\therefore \frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} \quad \therefore \text{eq is exact}$

There is a function $u(x, y)$ such that

$$\frac{\partial u}{\partial x} = M = x^2 - y^2 \quad \frac{\partial u}{\partial y} = N = -2xy \rightarrow (1)$$

$$u = \int M dx = \int (x^2 - y^2) dx = \frac{x^3}{3} - xy^2 + \phi(y) \rightarrow (2)$$

$$\therefore u(x, y) = \frac{x^3}{3} - xy^2 + \phi(y)$$

diff w.r. to y

$$\frac{\partial u}{\partial y} = -2xy + \phi'(y) \quad \text{ie } N = -2xy + \phi'(y)$$

But $N = -2xy \quad \therefore \phi'(y) = 0 \quad \therefore \phi(y) = C$

Sub in (1) $u(x, y) = \frac{x^3}{3} - xy^2 + C$

$\therefore$ Soln of the given diff. eq is

$$u(x, y) = C_1$$

$$\text{ie } \frac{x^3}{3} - xy^2 + C = C_1 \Rightarrow \underline{\underline{\frac{x^3}{3} - xy^2 = C_1}}$$

Short-cut method

Solution is given by

$$\int M dx + \int (\text{terms of } N \text{ not containing } x) dy = C$$

Example (1) $(x^2 - y^2) dx - 2xy dy = 0$

$$M = x^2 - y^2 \quad N = -2xy$$

Soln: $\int (x^2 - y^2) dx + \int 0 dy = C$

$$\text{ie } \underline{\underline{\frac{x^3}{3} - xy^2 = C}}$$

(2) Solve $(3x^2 + 4xy) dx + (2x^2 + 2y) dy = 0$

$$M = 3x^2 + 4xy, \quad N = 2x^2 + 2y$$

$$\frac{\partial M}{\partial y} = 4x, \quad \frac{\partial N}{\partial x} = 4x$$

$$\therefore \frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} \quad \therefore \text{Eq is exact}$$

Soln: $\int M dx + \int (\text{terms of } N \text{ not containing } x) dy = C$

$$\text{ie } \int (3x^2 + 4xy) dx + \int 2y dy = C$$

$$\text{ie } \frac{3x^3}{3} + 4 \frac{x^2 y}{2} + y^2 = C$$

$$\text{ie } x^3 + 2x^2 y + y^2 = C$$

Solve the following diff eqs

$$(1) (3x^2 + 4xy) dx + (2x^2 + 2y) dy = 0$$

$$(2) (2x \cos y + 3x^2 y) dx + (x^3 - x^2 \sin y - y) dy = 0$$

$$(3) (5x^4 + 3x^2 y^2 - 2xy^3) dx + (2x^3 y - 3x^2 y^2 - 5y^4) dy = 0$$

$$(4) (2xy + y - \tan y) dx + (x^2 - x \tan^2 y + \sec^2 y + 2) dy = 0$$

$$(2) (2x \cos y + 3x^2 y) dx + (x^3 - x^2 \sin y - y) dy = 0$$

$$M = 2x \cos y + 3x^2 y, \quad N = x^3 - x^2 \sin y - y$$

$$\frac{\partial M}{\partial y} = -2x \sin y + 3x^2, \quad \frac{\partial N}{\partial x} = 3x^2 - 2x \sin y$$

$$\therefore \frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} \text{ Hence eq. is exact}$$

Solution is given by

$$\int M dx + \int (\text{terms of } N \text{ not containing } x) dy = C$$

$$\therefore \int (2x \cos y + 3x^2 y) dx + \int (-y) dy = C$$

$$\therefore x^2 \cos y + x^3 y - \frac{y^2}{2} = C$$

$$(3) (5x^4 + 3x^2 y^2 - 2xy^3) dx + (2x^3 y - 3x^2 y^2 - 5y^4) dy = 0$$

$$M = 5x^4 + 3x^2 y^2 - 2xy^3, \quad N = 2x^3 y - 3x^2 y^2 - 5y^4$$

$$\frac{\partial M}{\partial y} = 6x^2 y - 6xy^2, \quad \frac{\partial N}{\partial x} = 6x^2 y - 6xy^2$$

$$\therefore \frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} \text{ Hence eq. is exact}$$

Solution is given by

$$\int M dx + \int (\text{terms of } N \text{ not containing } x) dy = C$$

$$\therefore \int (5x^4 + 3x^2 y^2 - 2xy^3) dx + \int (-5y^4) dy = C$$

$$\therefore x^5 + x^3 y^2 - x^2 y^3 - y^5 = C$$

$$(4) (2xy + y - \tan y) dx + (x^2 - x \tan^2 y + \sec^2 y + 2) dy = 0$$

$$M = 2xy + y - \tan y, \quad N = x^2 - x \tan^2 y + \sec^2 y + 2$$

$$\frac{\partial M}{\partial y} = 2x + 1 - \sec^2 y$$

$$= 2x + 1 - (1 + \tan^2 y)$$

$$= \underline{2x - \tan^2 y}$$

$$\frac{\partial N}{\partial x} = \underline{2x - \tan^2 y}$$

$\therefore \frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$. Hence eq is exact

Soln is given by

$$\int M dx + \int (\text{terms of } N \text{ not containing } x) dy = C$$

$$\text{i.e. } \int (2xy + y - \tan y) dx + \int (\sec^2 y + 2) dy = C$$

$$\text{i.e. } x^2 y + xy - x \tan y + \tan y + 2y = C$$

$$\text{i.e. } x^2 y + xy + (1-x) \tan y + 2y = C$$

(5) Under what condition the diff. eq

$$(ax + by) dx + (lx + my) dy = 0 \text{ is exact?}$$

Solve the resulting diff. eq.

$$M = ax + by, \quad N = lx + my$$

$$\frac{\partial M}{\partial y} = b, \quad \frac{\partial N}{\partial x} = l$$

The eq is exact if and only if

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} \Rightarrow b = l \text{ is the required condition. Resulting diff eq is}$$

$$(ax + by) dx + (bx + my) dy = 0 \text{ which is exact}$$

Solution is given by

$$\int (ax + by) dx + \int my dy = 0$$

$$\text{i.e. } a \frac{x^2}{2} + bxy + \frac{my^2}{2} = \frac{C}{2} \Rightarrow ax^2 + 2bxy + my^2 = C$$

(6) Test the exactness and solve the initial value problem

$$ye^x dx + (2y + e^x) dy = 0, \quad y(0) = -1$$

$$M = ye^x \quad N = 2y + e^x$$

$$\frac{\partial M}{\partial y} = e^x, \quad \frac{\partial N}{\partial x} = e^x$$

$\therefore \frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$ Hence the eq is exact

Solution is given by

$$\int M dx + \int (\text{terms of } N \text{ not involving } x) dy = C$$

$$\therefore \int ye^x dx + \int 2y dy = C$$

$$\therefore ye^x + y^2 = C \rightarrow (1) \text{ is the general solution}$$

Given that $y(0) = -1$. This means that when $x=0$, $y=-1$

$\therefore$ From eq (1) we get,

$$-1 \cdot e^0 + 1 = C \Rightarrow C = -1 + 1 = 0$$

Required solution is given by

$$\underline{ye^x + y^2 = 0}$$

Solve the following diff. eqs.

$$(1) (y \cos x + x) dx + (\sin x + y) dy = 0$$

$$(2) (3x^2 + 6xy^2) dx + (6x^2y + 4y^3) dy = 0$$

$$(3) e^x dx + (xe^y + xy) dy = 0$$

$$(4) \frac{y}{x^2} dx + (y - \frac{1}{x}) dy = 0$$

Simple applications

slope at any point (x, y) of a curve is given by $1 + \frac{y}{x}$. If the curve passes through $(1, 1)$ find its equation

Given that $\frac{dy}{dx} = 1 + \frac{y}{x} \rightarrow (1)$

This is a homogeneous diff. eq

Put $y = vx$ $\therefore \frac{dy}{dx} = v + x \frac{dv}{dx}$

sub in (1) $v + x \frac{dv}{dx} = 1 + v$

$\therefore x \frac{dv}{dx} = 1 \Rightarrow dv = \frac{dx}{x}$

Integrating, $v = \log x + c$

$\therefore \frac{y}{x} = \log x + c \rightarrow (2)$

This curve passes through $(1, 1)$

$\therefore 1 = \log 1 + c \Rightarrow c = 1 \quad (\because \log 1 = 0)$

sub in (2), $\frac{y}{x} = \log x + 1$

$\therefore y = x(\log x + 1)$ is the required

Solution

(2) If the temperature of a cake is 300°F when it is taken from oven and 200°F ten minutes later. When will it be practically equal to the room temperature of 60°F ?

Solution According to Newton's law of cooling, rate of change of temperature T is proportional to difference between T and the temp. of the surroundings 60°F

$\therefore \frac{dT}{dt} = k(T - 60)$ where k is a constant

ie $\frac{dT}{dt} - KT = -60K$ This is a linear diff. eq.

$$I.F = e^{\int -K dt} = e^{-Kt}$$

general soln: $T(e^{-Kt}) = \int e^{-Kt} \cdot (-60K) dt$

$$T \cdot e^{-Kt} = -60K \frac{e^{-Kt}}{-K} + C$$

$$ie \quad T e^{-Kt} = 60 e^{-Kt} + C$$

$$\therefore T = 60 + C e^{Kt} \rightarrow (1)$$

Initially $T(0) = 300$ ie $T = 300$ when $t = 0$

$$\therefore 300 = C e^0 + 60$$

$$ie \quad C = 240$$

Sub in eq (1) $T = 60 + 240 e^{Kt} \rightarrow (2)$

Also $T(10) = 200$ ie $T = 200$ when $t = 10$

$$\therefore 200 = 60 + 240 e^{10K}$$

$$\Rightarrow 140 = 240 e^{10K} \Rightarrow \frac{140}{240} = e^{10K}$$

$$\Rightarrow \frac{7}{12} = e^{10K} \Rightarrow 10K = \log\left(\frac{7}{12}\right)$$

$$\Rightarrow K = \frac{\log\left(\frac{7}{12}\right)}{10} = -0.0539$$

Sub in eq (2) $T = 60 + 240 e^{-(0.539)t}$
This is the temp. of the cake at any time t.